

DBMS LAB(31-08-2026)

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JOIN SQL COMMANDS

CREATING STUDENT AND DEPARTMENT TABLES:

The image shows two screenshots of a SQL Worksheet interface. The first screenshot displays the SQL commands to create the 'Student' and 'Department' tables. The 'Student' table has columns 'student_id' (INT PRIMARY KEY), 'student_name' (VARCHAR(50)), and 'dept_id' (INT). The 'Department' table has columns 'dept_id' (INT PRIMARY KEY) and 'dept_name' (VARCHAR(50)). The second screenshot shows the execution of these commands, with the 'Script output' tab selected. It displays the SQL commands and the resulting messages: 'Table STUDENT created.' and 'Table DEPARTMENT created.'

```
[ SQL Worksheet ]* 

1 CREATE TABLE Student (
2     student_id INT PRIMARY KEY,
3     student_name VARCHAR(50),
4     dept_id INT
5 );
6 CREATE TABLE Department (
7     dept_id INT PRIMARY KEY,
8     dept_name VARCHAR(50)
9 );

Query result  Script output  DBMS output  Explain Plan  SQL history



SQL> CREATE TABLE Student (
    student_id INT PRIMARY KEY,
    student_name VARCHAR(50),
    dept_id INT...
Show more...

Table STUDENT created.
Elapsed: 00:00:00.01s

[ SQL Worksheet ]* 

6 CREATE TABLE Department (
7     dept_id INT PRIMARY KEY,
8     dept_name VARCHAR(50)
9 );
10 INSERT INTO Student (student_id, student_name, dept_id)
11 VALUES
12 (101, 'Rahul', 10),
13 (102, 'Priya', 20),
14 (103, 'Arjun', 10);

Query result  Script output  DBMS output  Explain Plan  SQL history



SQL> CREATE TABLE Department (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50)
)

Table DEPARTMENT created.
```

INSERTING VALUES INTO STUDENT AND DEPARTMENT TABLES :

```
10 INSERT INTO Student (student_id, student_name, dept_id)
11 VALUES
12 (101, 'Rahul', 10),
13 (102, 'Priya', 20),
14 (103, 'Arjun', 10),
15 (104, 'Sneha', 30),
16 (105, 'Kiran', 40);
17 INSERT INTO Department (dept_id, dept_name)
```

Query result **Script output** DBMS output Explain Plan SQL

 

```
SQL> INSERT INTO Student (student_id, student_name, dept_id)
VALUES
(101, 'Rahul', 10),
(102, 'Priya', 20),...
show more...

4 rows inserted.
Elapsed: 00:00:00.142
```

```
17 INSERT INTO Department (dept_id, dept_name)
18 VALUES
19 (10, 'CSE'),
20 (20, 'ECE'),
21 (30, 'EEE'),
22 (50, 'MECH');
23 select student_id, student_name, dept_name from student, DEPARTMENT;
24
```

Query result **Script output** DBMS output Explain Plan SQL history

 

```
SQL> INSERT INTO Department (dept_id, dept_name)
VALUES
(10, 'CSE'),
(20, 'ECE'),...
show more...

4 rows inserted.
Elapsed: 00:00:00.058
```

CARTESIAN JOIN :

```

22 SELECT * FROM STUDENT;
23 SELECT student_id, student_name, dept_name FROM student, DEPARTMENT;
24
25 SELECT student.student_id,

```

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.002 seconds

	STUDENT_ID	STUDENT_NAME	DEPT_NAME
8	103	Arjun	ECE
9	104	Sneha	ECE
10	105	Kiran	ECE
11	101	Rahul	EEE
12	102	Priya	EEE
13	103	Arjun	EEE

SQL JOIN :

```

25 SELECT student.student_id,
26 student.student_name,
27 department.dept_id,
28 department.dept_name
29 FROM student, department
30 WHERE student.dept_id = department.dept_id;
31
32 SELECT * FROM student;

```

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.001 seconds

	STUDENT_ID	STUDENT_NAME	DEPT_ID	DEPT_NAME
1	101	Rahul	10	CSE
2	102	Priya	20	ECE
3	103	Arjun	10	CSE
4	104	Sneha	30	EEE

SELF JOIN:

```

22 SELECT * FROM STUDENT;
23 SELECT s1.student_id AS student1,
24 s2.student_id AS student2,
25 s1.dept_id AS dept1,
26 s2.dept_id AS dept2
27 FROM student s1, student s2
28 WHERE s1.dept_id = s2.dept_id;

```

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.008 seconds

	STUDENT1	STUDENT2	DEPT_ID
1	Arjun	Rahul	10
2	Priya	Priya	20
3	Rahul	Arjun	10
4	Arjun	Arjun	10
5	Sneha	Sneha	30

OUTER JOIN:

```

22 SELECT student_id, student_name, dept_id FROM student;
23 SELECT a.student_id,
24 a.student_name,
25 b.dept_id,
26 b.dept_name
27 FROM student a, department b
28 WHERE a.dept_id(+) = b.dept_id;

```

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.01 seconds

	STUDENT_ID	STUDENT_NAME	DEPT_ID	DEPT_NAME
1	101	Rahul	10	CSE
2	102	Priya	20	ECE
3	103	Arjun	10	CSE
4	104	Sneha	30	EEE
5	(null)	(null)	50	MECH